# Advancements in Phishing Website Detection: A Comprehensive Analysis of Machine Learning and Deep Learning Models

Soudabeh Mousavi
*Computer Engineering Department*
*Imam Khomeini International University*
Qazvin, Iran
soudabe.mousavi@edu.ikiu.ac.ir

Mahdi Bahaghighat *
*Computer Engineering Department*
*Imam Khomeini International University*
Qazvin, Iran, *Corresponding author
Bahaghighat@eng.ikiu.ac.ir

Figen Ozen
*Halic University*
İstanbul, Türkiye
figenozen@halic.edu.tr

2024 32nd Signal Processing and Communications Applications Conference (SIU) | 979-8-3503-8896-1/24/$31.00 ©2024 IEEE | DOI: 10.1109/SIU61531.2024.10601036

***Abstract*— Today, machine learning (ML) and deep learning (DL) have become potent tools for detecting phishing websites. This paper has employed a bunch of ML algorithms such as Random Forest, Extra Tree, XGBoost, LGBM, and XGBoost-bagging, along with some proposed neural network architectures found on Multilayer Perceptron (MLP), Convolutional Neural Network (CNN), and Long Short-Term Memory (LSTM) for identifying website phishing. Implementing a comprehensive approach, we devised our model based on 56 proposed features extracted from each target website using Feature Importance. Then, we conducted five strategic steps to enhance the performance of ML and DL models. Through these steps, we improved DL model accuracy remarkably, notably elevating LSTM accuracy from the lowest accuracy of 96.18% to 98.78% with an AUC-ROC LSTM of about 99.85%.**

***Keywords*— *Phishing website detection, CNN, LSTM, MLP, Machine Learning, Deep Learning***

## I. Introduction

Today, phishing is a sophisticated cyber-attack usually combined with social engineering methods. Hackers utilize users' instincts, curiosity, trust, fear, and greed to commit this cyber-attack. They do not need to crack complex encryption codes or bypass a strict firewall. Such attacks have different forms and usually include communication channels such as email, instant messages, and social media. Regardless of the channel used, attackers often impersonate well-known banks, credit card companies, or e-commerce websites to encourage users to click on a hyperlink embedded in the body of an email or text message sent by the attacker. Then, by that link, users are directed to fake web pages similar to the targeted legitimate websites, and their confidential and personal information is exposed and at risk of being revealed or stolen. Diverse communication channels can send fake links [1]. Due to many validated reports, phishing has increased since it emerged [2]. There are multiple reasons for this attack's growth. Firstly, online users are unaware of this kind of attack. However, experienced users can also get trapped in phishing attacks. Subsequently, hackers update their methods and use new technologies [3]. Unfortunately, most available techniques to confront phishing are impractical [4]. Now, machine learning and deep learning are used alone or as hybrid techniques in various applications [5-6]. Feature engineering is a critical procedure in ML because attackers may change the features they use due to technological advancements. However, phishing websites may also use features like SSL or HTTPS to mimic legitimate ones [7]. Therefore, it is necessary to scrutinize features to tackle phishing detection. [8] focused on ML algorithms like RF, C4.5, and PART for website phishing detection. It utilized around 48 traditional features. Feature selection was performed using hybrid ensemble techniques. The best-performing model was the RF, which achieved an accuracy rate of 94.6%. Authors in [9] employed various ML algorithms such as RF, Decision Tree, K-NN, SVM, and Naïve-Bayes. They conducted a wrapper-based technique for Feature selection alongside the MillerSmiles [10] and PhishTank [11] datasets. However, the dataset size and the ratio between phishing and legitimate instances were unspecified. The best accuracy rate was RF at 96.87%. [12] aimed to use a CNN model to identify phishing using the PhishTank [11] dataset.

In this paper, we widely used ML and DL techniques and proposed a five-step approach to improve the performance of models in phishing detection. A comprehensive analysis of models and a selection of 56 key affective features have been conducted to get the most accurate model. Firstly, because every model has various hyperparameters, tuning them as a first step is essential. Then, the number of features decreased by the Importance Feature method, and in the next step, all ML models rerun with these features. In the subsequent step, the DL models were designed. Finally, the segmentation of data is transformed.

The rest of the paper is organized as follows: Section II presents our work's methodology. Section III discusses the experiment results. Ultimately, the final section describes the conclusion of the work.

## II. Methodology

Phishing website detection is a crucial task and a binary classification problem. The aim is to explore how ML and DL techniques can be employed to identify phishing websites. This paper utilizes a bunch of solid algorithms like Random Forest (RF), Extra Tree (ET), eXtreme Gradient Boosting (XGB), Light Gradient Boosted Machines (LGBM), and XGB-bagging to achieve this aim. Besides, we designed and optimized several Neural Networks based on learning algorithms like Convolutional Neural Networks (CNN), Multilayer Perceptron

979-8-3503-8896-1/24/$31.00 ©2024 IEEE

orized licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on August 27,2024 at 10:12:10 UTC from IEEE Xplore. Restrictions a

(MLP), and Long Short-Term Memory (LSTM) to find the best solution for spotting phishing websites.

### A. The proposed steps to detect phishing websites

We offer five steps to enhance models' performance. The description of the five steps is as follows.

- **Step 1:** After preprocessing the data, ML model architectures are proposed. The GrideSearchCV function is used to tune hyperparameters. The architectures are elaborated with full details in Table 1.
- **Step 2:** We aim to reduce features and simultaneously observe the effect on ML models' behavior afterward. Importance Features method is a dimension reduction technique used in each ML algorithm to find the most key features. Lastly, all prominent features were unified.
- **Step 3:** ML models are rerun with 56 features selected in the previous step to monitor their behavior separately.
- **Step 4:** Key features in Step 2 are used here to tune DL models' hyperparameters using the RandomSearch [13] in Keras. Table 2 shows the architecture of DL models.
- **Step 5:** Conventional methods of splitting data were employed in earlier steps. The ratio was 80:20. However, Stratified K-fold cross-validation is used with a value of k=10 in the fifth step. This change aims to strengthen the reliability of the phishing detection model. This shift to cross-validation is due to ensuring a thorough assessment of the model's effectiveness against unseen data.

All model results in every step are compared with each other to gain better insight into their behavior through steps to detect website phishing.

### B. Dataset and Data Preprocessing

This paper used a public dataset presented in [14]. It has 111 features with one label for prediction, is unbalanced, and has 88,000 samples. Only 59 key features were picked as prominent features to detect phishing. Data normalization is an essential stage in preprocessing. The best normalization method choice depends on the dataset's timing and characteristics. Experience and testing can efficiently help to choose suitable method [15]. In this work, the StandardScaler is used to normalize the data. As mentioned earlier, our dataset is imbalanced. Therefore, it is necessary to balance it. The Synthetic Minority Oversampling Technique (SMOTE) [16] is employed to increase the number of classes with minority samples. The normalization and balancing dataset techniques in all steps of this work are the same.

TABLE 1. FINAL DESIGN FOR PROPOSED ML MODELS

| ML Model | Architecture |
|---|---|
| Random Forest | bootstrap = False, max_depth = 30, max_features = 'sqrt', n_estimators = 200 |
| ExtraTree | bootstrap = True, min_samples_spli t= 4, max_features = None, n_estimators = 100 max_depth = 20 |
| XGBoost | n_estimators = 400, max_depth =11, learning_rate = 1, objective = 'binary:logistic' |
| LightGBM | learning_rate = 0.1, max_depth = 10, n_estimators = 300, num_leaves = 63 |
| XGB-Bagging | base_model = XGBoost, |

### C. Machine Learning Classifiers

This paper uses Random Forest, Extra Tree, XGB, LGBM, and XGB-bagging to detect website phishing. Random Forest (RF) is a famous Ensemble Learning Algorithm that combines several decision trees (DT) for prediction [17]. Extremely random trees (ET) is another ML algorithm that comprises decision tree collections related to other groups of DT algorithms, such as bootstrap clustering and RF [18]. EXtreme Gradient Boosting (XGB) is an open-source gradient boosting framework. It works by building a set of DT. Light Gradient Boosted Machines (LGBM) is an open-source gradient boosting framework initially developed and published by Microsoft. Bagging is a short-term bootstrap aggregating. This name refers to how the ensemble diversity is achieved (through bootstrap sampling) and ensemble prediction is performed [19].

### D. Deep Learning Classifiers

Sometimes, ML algorithms fail to find proper solutions for more complex problems. As a result, DL techniques can be used to identify phishing. In this study, the Convolutional Neural Network (CNN), Long Short-Term Memory (LSTM), and Multilayer perceptron (MLP) are conducted. CNN is a neural network with multiple layers used in various fields, such as image processing and recognition of handwritten characters. LSTM uses a memory unit called a cell, which can maintain its value for a sufficient period and behave accordingly. MLP is a widely used model in neural networks. It has a perceptron with a non-linear activation function and several different layers [20].

### E. Evaluation Metrics

The model performance needs to be evaluated, so there are many evaluation metrics for specific problems. The confusion matrix summarizes the model's prediction in binary classification problems [21]. The other essential metrics for phishing detection are False Positive Rate (FPR), False Negative Rate (FNR), Accuracy, Precision, Recall, and f1 Score. All of these metrics' equations are presented in Table 3. This paper also employs Area under the ROC Curve (AUC-ROC) [22].

TABLE 2. FINAL DESIGN FOR PROPOSED DL MODELS

| Model | Architecture |
|---|---|
| MLP | Input = 56, hidden layer =256, 128, 64, activation function = ReLU, Dropout Layers = 0.1 (after each hidden layer), Kernel Initialization = normal, Output Layer = Sigmoid, Optimizer = Admax, learning_rate = 0.001, epoch = 70, batch_size = 32 |
| CNN | Input = (None, 1, 56), Convolutional Layers = 100 (kernel size 60)/ 64 (kernel size 20)/ 75 (kernel size 10), Activation Function = ReLU, Dropout Layers = 0.1, MaxPooling1D = 7, Fully Connected Layers = Dense layer with 256,128,64, units and ReLU activation, Optimizer = Adamax, Learning_rate = 0.001, epoch = 100, batch_size =32 |
| LSTM | Input = (None, 1, 56), hidden layer = 256, 128, 64, activation function = ReLU, Dropout Layers = 0.1 (after each LSTM layer), LSTM Return Sequences (True (for first two LSTMs)), Output Layer = Sigmoid, Optimizer = Admax, learning_rate = 0.001, batch_size = 36, epoch = 40 |

orized licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on August 27,2024 at 10:12:10 UTC from IEEE Xplore. Restrictions a

### F. Environment

This work used Python's state-of-the-art data science tools to implement all models. The implemented codes were run on a personal computer with an Intel Xeon e5-1620 processor and an RX 580 Radeon 8GB graphics card with 32GB of RAM.

## III. Experimental Results

Our study offers a five-step approach to enhancing models' performance. In Step 1, ML model architectures were designed and optimized based on 59 features. The results are presented in Table 4. Each model demonstrates strong classification capabilities and high ACC, precision, recall, and F1 scores, ranging from 97.52% to 98%. These results suggest that models can precisely detect phishing websites. Dimension reduction was implemented in Step 2. Thus, 56 features were introduced. Fig. 1 presents these key features. In Step 3, we aim to evaluate ML models' behavior after dimension reduction. So, all ML Models were rerun using 56 features extracted in Step 2.

TABLE 3. EVALUATION METRICS FOR MODEL ANALYSIS

| Metrics | Equation |
|---|---|
| False Positive Rate | $FPR=\frac{FP}{(FP+TN)}$ |
| False Negative Rate | $FNR=\frac{FN}{(FN+TP)}$ |
| Accuracy | $ACC=\frac{TP+TN}{(TP+FP+FN+TN)}$ |
| Precision | $Precision=\frac{TP}{(TP+FP)}$ |
| Recall | $Recall=\frac{TP}{(TP+FN)}$ |
| f1 Score | $F1Score=\frac{2*(Precision*Recall)}{(Precision+Recall)}$ |

TABLE 4. STEP 1 RESULTS (59 FEATURES)

| Models | Step 1 – ML Design | | | | | |
|---|---|---|---|---|---|---|
| | *Acc* | *Precision* | *Recall* | *F1* | *FPR* | *FNR* |
| RF | 97.91% | 97.54% | 98.25% | 97.89% | 2.4% | 1.7% |
| ET | 97.52% | 97.05% | 97.97% | 97.51% | 2.9% | 2.0% |
| XGB | 97.85% | 97.71% | 97.96% | 97.83% | 2.2% | 2.0% |
| LGBM | 98.00% | 97.88% | 98.08% | 97.98% | 2.8% | 1.9% |
| XGB-b | 98.00% | 97.83% | 98.14% | 97.98% | 2.1% | 1.8% |

Fig. 1. 56 selected features by Importance Feature

TABLE 5. ML MODEL RESULTS AFTER DIMENSION REDUCTION (56 FEATURES)

| Models | Step 3 – After dimension Reduction | | | | | |
|---|---|---|---|---|---|---|
| | *Acc* | *Precision* | *Recall* | *F1* | *FPR* | *FNR* |
| RF | 97.89% | 97.30% | 98.48% | 97.89% | 2.6% | 1.5% |
| ET | 97.48% | 96.76% | 98.22% | 97.48% | 3.2% | 1.7% |
| XGB | 97.88% | 97.50% | 98.24% | 97.87% | 2.4% | 1.7% |
| LGBM | **98.04%** | **97.65%** | **98.48%** | **98.03%** | **2.3%** | **1.5%** |
| XGB-b | 98% | 97.47% | 98.51% | 97.99% | 2.5% | 1.4% |

As seen in Table 5, the models' performance after dimension reduction reveals subtle but noticeable shifts in their metrics. However, the AUC-ROC scores also remarkably improved for all models, resulting in exceptional scores of 99.80%**.** In Step 4, DL models were designed. Table 6 shows the models' performance. As can be seen, despite DL's good results, they fell behind ML models. Step 5 is a different data segmentation method used to enhance the models' performance, and Table 7 demonstrates the experiment results. The outcomes were incredible for DL models—their performance yields. However, ML models' ACC marginally improved except for LGBM. As Table 7 presents, after following five steps, XGB-b performs higher in the ML and has reached 98.04% ACC and LSTM with 98.78% ACC. The AUC-ROC scores for XGB-b and LSTM are 99.80% and 99.85%, respectively. Results show that applying stratified k-fold has boosted performance across models to detect phishing, especially for DL ones.

TABLE 6. DL RESULTS AFTER DIMENSION REDUCTION (56 FEATURES)

| Models | Step 4 – DL Design | | | | | |
|---|---|---|---|---|---|---|
| | *Acc* | *P* | *Recall* | *F1* | *FPR* | *FNR* |
| MLP | 97.07% | 96.65% | 97.46% | 97.06% | 3.2% | 2.5% |
| CNN | **97.07%** | **97.16%** | **96.91%** | **97.03%** | **1.4%** | **2.4%** |
| LSTM | 96.18% | 96.59% | 96.65% | 96.12% | 3.3% | 4.3% |

TABLE 7. COMPARING ALL RESULTS TOGETHER

| Models | Step 5 – Comparing ML and DL results | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | *MLP* | *CNN* | *LSTM* | *RF* | *ET* | *XGB* | *LGBM* | *XGB-b* |
| Accuracy | 98.4 | 98.01 | **98.78** | 97.96 | 97.49 | 97.95 | 97.95 | 98.04 |
| Precision | 98.5 | 98.49 | **98.88** | 97.60 | 97.05 | 97.80 | 97.84 | 97.81 |
| Recall | 98.2 | 97.51 | **98.68** | 98.34 | 97.96 | 98.11 | 98.05 | 98.29 |
| F1 Score | 98.4 | 98 | **98.78** | 97.97 | 97.50 | 97.95 | 97.95 | 98.05 |
| AUC-ROC | 99.8 | 99.78 | **99.85** | 99.80 | 99.80 | 99.80 | 99.80 | 99.80 |
| FPR | 1.4 | 1.4 | **1.1** | 2.4 | 2.9 | 2.2 | 2.1 | 1.7 |
| FNR | 1.7 | 2.4 | **1.3** | 1.6 | 2.0 | 1.8 | 1.9 | 1.7 |

Finally, Table 8 shows our work compared with other available studies. Accordingly, this study could gain higher accuracy, with the proposed LSTM achieving 98.78% and MLP achieving 98.43%, while [23] delivered for MLP with about 95.27% accuracy. Other experiments, such as [24] and [25], reached 96.8% and 93.45% ACC for LSTM, respectively.

orized licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on August 27,2024 at 10:12:10 UTC from IEEE Xplore. Restrictions a

TABLE 8. COMPARED WITH OTHER STUDIES

| Model | *LSTM* | *MLP* | *No. Features* |
|---|---|---|---|
| Proposed | 98.78% | 98.43% | 56 |
| [23] | - | 95.27% | 9 |
| [24] | 96.80% | - | 30 |
| [25] | 93.45% | - | 9 |

## IV. CONCLUSION

Concerted efforts and innovative solutions are necessary to mitigate the threats of phishing attacks and prevent potential losses in the digital world. Such work builds on leveraging ML and DL techniques to strengthen cyber defenses. Significant advancement is noted in our proposed five-step approach from feature selection to refined data segmentation. In the first step, the designed ML models showed great promise as most of the algorithms were measured at nearly 98% accuracy. More importantly, LGBM and XGB-b performed excellently, with an AUC-ROC of 99.77%. Feature dimension reduction is the second step, which reduces the feature set to 56, but the change in ML models' behavior is attributed except for XGB-b, which recorded the same AUC-ROC of 99.80% for all models. The implementation of DL models witnessed a 97.07% accuracy for CNN and MLP, with a little lagging LSTM recording 96.18% accuracy. However, the final step implementation of stratified k-fold validation with LSTM took the lead with an outstanding score of 98.78% accuracy and AUC-ROC 99.85%, superseding the collective performance of other models for detecting phishing websites. These five steps represent enormous gains in model efficiency and pave the way to strengthen cyber defenses from the growing threat of phishing websites.

## REFERENCES


[1] M. Alqahtani, "Phishing Websites Classification using Association Classification (PWCAC)," in International Conference on Computer and Information Sciences 2019 (ICCIS), 3-4 April 2019, pp. 1-6, doi: https://doi.org/10.1109/ICCISci.2019.8716444.

[2] A. P. W. Group, " The APWG Phishing Activity Trends Report," Report analyse Nov.02 2023. [Online]. Available: https://apwg.org/trendsreports/

[3] M. Volkamer, K. Renaud, B. Reinheimer, and A. Kunz, "User experiences of TORPEDO: TOoltip-poweRed Phishing Email DetectiOn," Computers & Security, vol. 71, no. 0167-4048, pp. 100–113, Nov. 2017, doi: https://doi.org/10.1016/j.cose.2017.02.004.

[4] A. K. Jain and B. B. Gupta, "A survey of phishing attack techniques, defence mechanisms and open research challenges," Enterprise Information Systems, vol. 16, no. 4, pp. 1–39, Mar. 2021, doi: https://doi.org/10.1080/17517575.2021.1896786.

[5] E. Amouee, Morteza Mohammadi Zanjireh, Mahdi Bahaghighat, and M. Ghorbani, "A new anomalous text detection approach using unsupervised methods," Facta Universitatis. Series Electronics and Energetics/Facta Universitatis. Series: Electronics and Energetics, vol. 33, no. 4, pp. 631–653, Jan. 2020, doi: https://doi.org/10.2298/fuee2004631a.

[6] G. Vrbančič, I. Fister, and V. Podgorelec, "Swarm Intelligence Approaches for Parameter Setting of Deep Learning Neural Network: Case Study on Phishing Websites Classification," presented at the Proceedings of the 8th International Conference on Web Intelligence, Mining and Semantics, Novi Sad, Serbia, 2018. [Online]. Available: https://doi.org/10.1145/3227609.3227655.

[7] P. Singh, N. Jain, and A. Maini, "Investigating the effect of feature and dimensionality reduction on phishing website classification problem," in *2015 1st International Conference on Next Generation Computing Technologies (NGCT)*, 4-5 Sept. 2015 2015, pp. 388-393, https://doi.org/10.1109/NGCT.2015.7375147.

[8] K. L. Chiew, C. L. Tan, K. Wong, K. S. C. Yong, and W. K. Tiong, "A new hybrid ensemble feature selection framework for machine learning-based phishing detection system," Information Sciences, vol. 484, no. 0020-0255, pp. 153–166, May 2019, doi: 10.1016/j.ins.2019.01.064.

[9] G. Harinahalli Lokesh and G. BoreGowda, "Phishing website detection based on effective machine learning approach," Journal of Cyber Security Technology, vol. 5, no. 1, pp. 1-14, 2021/01/02 2021, doi: 10.1080/23742917.2020.1813396.

[10] MillerSmiles. "Phishing scams and spoof emails at MillerSmiles.co.uk." http://www.millersmiles.co.uk/ (accessed Feb. 20, 2024).

[11] D. Ulevitch. "PhishTank | Join the fight against phishing." https://phishtank.org/ (accessed Feb. 20, 2024).

[12] N. Al-Milli and B. H. Hammo, "A Convolutional Neural Network Model to Detect Illegitimate URLs," in 2020 11th International Conference on Information and Communication Systems (ICICS), Irbid, Jordan: IEEE, Apr. 2020, pp. 220–225. doi: 10.1109/ICICS49469.2020.239536

[13] K. Team, "Keras documentation: RandomSearch Tuner," *keras.io*. [Online]. Available: https://keras.io/api/keras_tuner/tuners/random/.

[14] G. Vrbančič, I. Fister, and V. Podgorelec, "Datasets for phishing websites detection," *Data in Brief,* vol. 33, p. 106438, 2020/12/01/ 2020, doi: 10.1016/j.dib.2020.106438.

[15] Kelsy Cabello-Solorzano, O. de, M. A. Peña, L. Correia, and A. J. Tallón-Ballesteros, "The Impact of Data Normalization on the Accuracy of Machine Learning Algorithms: A Comparative Analysis," *Lecture notes in networks and systems*, vol. 750, pp. 344–353, Jan. 2023, doi: https://doi.org/10.1007/978-3-031-42536-3_33

[16] D. Elreedy and A. F. Atiya, "A Comprehensive Analysis of Synthetic Minority Oversampling Technique for handling class imbalance," *Information Sciences,* vol. 505, pp. 32-64, 2019/12/01/ 2019, doi: 10.1016/j.ins.2019.07.070.

[17] M. Bahaghighat, M. Ghasemi, and F. Ozen, "A high-accuracy phishing website detection method based on machine learning," *Journal of Information Security and Applications,* vol. 77, p. 103553, 09/01 2023, doi: https://doi.org/10.1016/j.jisa.2023.103553.

[18] P. Geurts, D. Ernst, and L. Wehenkel, "Extremely randomized trees," *Machine Learning,* vol. 63, no. 1, pp. 3-42, 2006/04/01 2006, doi: https://doi.org/10.1007/s10994-006-6226-1.

[19] G. KUNAPULI, *Ensemble Methods for Machine Learning Book*, first ed. United States of America: Manning Publications Co., 2023, p. 354.

[20] S. Dargan, M. Kumar, M. R. Ayyagari, and G. Kumar, "A Survey of Deep Learning and Its Applications: A New Paradigm to Machine Learning," *Archives of Computational Methods in Engineering,* vol. 27, no. 4, pp. 1071-1092, 2020/09/01 2020, doi: 10.1007/s11831-019-09344-w.

[21] G. Varoquaux and O. Colliot, "Evaluating Machine Learning Models and Their Diagnostic Value," in *Machine Learning for Brain Disorders*, O. Colliot Ed. New York, NY: Springer US, 2023, vol. 197, pp. 601-630, https://doi.org/10.1007/978-1-0716-3195-9_20

[22] J. Davis and M. Goadrich, "The relationship between Precision-Recall and ROC curves," presented at the Proceedings of the 23rd international conference on Machine learning, Pittsburgh, Pennsylvania, USA, 2006. [Online]. Available: https://doi.org/10.1145/1143844.1143874.

[23] Priya Saravanan and S. Subramanian, "A Framework for Detecting Phishing Websites using GA based Feature Selection and ARTMAP based Website Classification," *Procedia Computer Science,* vol. 171, pp. 1083-1092, 2020. Elsevier, doi: 10.1016/j.procs.2020.04.116.

[24] Z. Alshingiti, R. Alaqel, J. Al-Muhtadi, Q. E. U. Haq, K. Saleem, and M. H. Faheem, "A Deep Learning-Based Phishing Detection System Using CNN, LSTM, and LSTM-CNN," *Electronics,* vol. 12, no. 1, p. 232, 2023. [Online]. Available: https://www.mdpi.com/2079-9292/12/1/232.

[25] Barah Mohammed Alanzi and a. D. M. Uliyan, "Detection of Phishing Websites by Investigating Their URLs using LSTM Algorithm," *IJCSNS International Journal of Computer Science and Network Security,* vol. 22, May 2022, doi: 10.22937/IJCSNS.2022.22.5.60.


orized licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on August 27,2024 at 10:12:10 UTC from IEEE Xplore. Restrictions a